

Solution Requirements

Date	25 June 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No authentication required; open-access web application for immediate student assistance. Future versions may include secure login for personalized learning history.	Low
2	Emotion Detection	Analyze student text using a dual-model approach (Machine Learning + Keyword Analysis) and classify emotions into Bored, Confident, Confused, Curious and Frustrated.	High
3	External Interfaces	Integrate Groq REST API (Llama 3.1 8B) to generate personalized learning guidance based on detected emotions.	High
4	Transaction Processing	User submits learning queries. Backend detects emotion, sends prompts to the Groq API and displays AI-generated responses in real time.	High
5	Reporting & Analytics	Store interaction history and display Plotly analytics to review emotional trends.	High
6	Business Rules	If API connectivity fails, automatically generate predefined empathetic responses based on detected emotion.	High
7	Compliance	No sensitive personal information is collected. Public datasets are used for model training.	Low
8	Other	Support future speech-to-text, cloud database integration and adaptive learning recommendations.	Medium

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Emotion detection and AI response generation should execute quickly after submission.	Boot < 1 second; AI response < 2 seconds
2	Scalability	Support multiple concurrent student sessions.	100+ concurrent users
3	Security & Data Privacy	Store API keys in environment variables; no sensitive data retained.	No hardcoded API keys

4	Reliability & Availability	Operate continuously without crashes.	99% uptime
5	Usability & Accessibility	Responsive, mobile-friendly interface requiring minimal effort.	Workflow within 3 clicks
6	Maintainability	Modular reusable code with clear documentation.	Easy module replacement